Brooke Jin

PhD Student, Brown University Department of Sociology
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Education

- **Brown University** **Expected 2031**
  Doctor of Philosophy in Sociology **Providence, RI**

- **Massachusetts Institute of Technology (MIT)** **September 2023 - May 2025**
  Master in City Planning **Cambridge, MA**

  Thesis title: Toward a political economy of the power sector: green capitalism, eco-socialism, and co-operative power in decarbonized climate policy
  Advisor: J. Phillip Thompson, Professor of Political Science and Urban Planning, MIT

- **Cornell University** **August 2017 - December 2020**
  Bachelor of Science in Mechanical Engineering (summa cum laude) **Ithaca, NY**

Peer-Reviewed Publications

- Janelle Knox-Hayes, Amanda Bendixen, Nikita Kulkarni, Nolen Scruggs, Shekhar Chandra, Jungwoo Chun, and **Brooke Jin**. "Mapping the Regional Ecologies of US Community-Based Organizations: Implications for Equitable Solar Energy Development." *Energy Policy* (2026), 220(115621). DOI: 10.1016/j.enpol.2026.115621.

- Jungwoo Chun, Dania Ortiz, **Brooke Jin**, Stephen Hart, Nikita Kulkarni, and Janelle Knox-Hayes. "Energy Burden in the United States: an Analysis Using Decision Trees." *Energies* (2025), 18(3), 646. DOI: 10.3390/en18030646.

- Zhiling Zheng, Federico Florit, **Brooke Jin**, Haoyang Wu, Shih-Cheng Li, Kakasaheb Y. Nandiwale, Chase A. Salazar, Jason G. Mustakis, William H. Green, and Klavs F. Jensen. "Integrating machine learning and large language models to advance exploration of electrochemical reactions." *Angew. Chem. Int. Ed.* (2024), e202418074. DOI: 10.1002/anie.202418074.

- Brent A. Koscher*, Richard B. Canty*, Matthew A. McDonald*, Kevin P. Greenman, Charles J. McGill, Camille L. Bilodeau, Wengong Jin, Haoyang Wu, Florence H. Vermeire, **Brooke Jin**, Travis Hart, Timothy Kulesza, Shih-Cheng Li, Tommi S. Jaakola, Regina Barzilay, Rafael Gomez-Bombarelli, William H. Green, and Klavs F. Jensen. "Autonomous, multiproperty-driven molecular discovery: From predictions to measurements and back." *Science* (2023), 382(6677). DOI: 10.1126/science.adi1407.

- Natalie S. Eyke, Timo N. Schneider, **Brooke Jin**, Travis Hart, Sebastien Monfette, Joel M. Hawkins, Peter D. Morse, Roger M. Howard, David M. Pfisterer, Kakasaheb Y. Nandiwale, and Klavs F. Jensen. "Parallel multi-droplet platform for reaction kinetics and optimization." *Chemical Science* (2023), 14(33), 8798−8809. DOI: 10.1039/D3SC02082G.

Reports and White Papers

- Janelle Knox-Hayes, Amanda Bendixen, Nikita Sanjay Kulkarni, Nolen Scruggs, Shekhar Chandra, Jungwoo Chun, and **Brooke Jin**. Rep. *US Community-Based Organizations and Their Relationship to Solar Energy Development.* Clean Energy States Alliance, August 2024. https://www.cesa.org/wp-content/uploads/US-Community-Based-Organizations-and-Their-Relationship-to-Solar-Energy-Development.pdf.

Grants

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- Watson PhD Research and Travel Award ($500) – Watson School of International and Public Affairs, Brown University

Presentations and Workshops

- "Political economy and financial(ized) geographies in state energy planning in Nebraska" – open panel, Society for Social Studies of Science (October 2026)
- "Political economy and financial geographies in state energy planning in Nebraska" – Environmental Sociology Workshop, Brown University (February 2026)

Awards and Honors

- O. Robert Simha Prize (2024) – awarded by the Massachusetts Institute of Technology to undergraduate or graduate students in the School of Architecture and Planning for outstanding work in planning
- Frank O. Ellenwood Prize (2020) – awarded by Cornell University to the mechanical engineering student who ranked highest in courses in power engineering

Research Experience

- **Massachusetts Institute of Technology (MIT)**
  - Janelle Knox-Hayes & the Resilient Communities Lab – Department of Urban Studies and Planning (January 2024 - August 2025)
  - Klavs F. Jensen Lab – Department of Chemical Engineering (July 2021 - January 2024)

Professional Experience

- Planning Intern at the City of Pawtucket, RI (May 2026 - August 2026)
- Carbon-Free Electricity Intern at Rocky Mountain Institute (June 2024 - August 2024)
- Energy Policy Intern at Public Grid (January 2024 - April 2024)
- Associate Hardware Engineer at Princeton Identity (January 2021 - June 2021)